

Comparing Two Proportions

Presentation Outline

- 1 Variables and Data
- 2 Sampling Distribution for $\hat{p}_1 - \hat{p}_2$
- 3 Confidence Interval for $p_1 - p_2$

Variables and Population Proportions

Variables:

- Grouping Variable ($I = 2$ groups)
- Response Variable ($J = 2$ categories)
 - Categories = Success and Failure

Population Proportions:

- p_1 = proportion of success in population 1
- p_2 = proportion of success in population 2

Compare population proportions between two populations.

Data Table

Categorical Variable	Grouping Variable
Success	Population 1
Failure	Population 1
⋮	⋮
⋮	⋮
Failure	Population 1
Failure	Population 2
Success	Population 2
⋮	⋮
⋮	⋮
Success	Population 2

Data - Contingency Table

Group	Response Variable		Total
	Success	Failure	
1	Y_{11}	Y_{12}	n_1
2	Y_{21}	Y_{22}	n_2
Total	$Y_{.1}$	$Y_{.2}$	n

Sample Statistics

Sample 1:

$\hat{p}_1 = \frac{Y_{11}}{n_1}$ = proportion of successes in sample 1

n_1 = number of observations in sample 1

Sample 2:

$\hat{p}_2 = \frac{Y_{21}}{n_2}$ = proportion of successes in sample 2

n_2 = number of observations in sample 2

Inference for $p_1 - p_2$

Goal: Obtain an interval of reasonable values of $p_1 - p_2$ given the data.

Estimate $p_1 - p_2$ with $\hat{p}_1 - \hat{p}_2$

What is the sampling distribution of $\hat{p}_1 - \hat{p}_2$?

Sampling Distribution for $\hat{p}_1 - \hat{p}_2$

Check the following conditions:

- $n_1\hat{p}_1 \geq 10$ and $n_1(1 - \hat{p}_1) \geq 10$
- $n_2\hat{p}_2 \geq 10$ and $n_2(1 - \hat{p}_2) \geq 10$

If the conditions hold, then we have that the distribution of:

$$z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}}$$

is well-approximated by the standard Normal distribution.

Sampling Distribution for $\hat{p}_1 - \hat{p}_2$

From the sampling distribution of $\hat{p}_1 - \hat{p}_2$, we have:

$$P \left(-z_{\alpha/2} < \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} < z_{\alpha/2} \right) \approx 1 - \alpha$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentile from a standard Normal distribution.

Solving the inequality above for $p_1 - p_2$ gives:

Confidence Interval for $p_1 - p_2$

$$P(\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} * SE(\hat{p}_1 - \hat{p}_2) < p_1 - p_2 < \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} * SE(\hat{p}_1 - \hat{p}_2)) \approx 1 - \alpha$$

$$\text{where } SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

An approximate $(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ is:

$$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} * SE(\hat{p}_1 - \hat{p}_2)$$

Example - Confidence Interval for $p_1 - p_2$

Obtaining a high response rate for surveys is very important to the overall validity of the results from a random sample of the population. A survey was administered to a random sample of around 10,000 doctors. Approximately half of the doctors were randomly selected to receive a letter before the survey. Researchers wanted to see if doctors who received the letter before the survey would respond to the survey at a different rate than doctors who did not receive the letter.

Variables:

- Grouping Variable: Received Letter, Did not Receive Letter
- Response Variable: Returned Survey (Success), Did not Return Survey (Failure)

Example - Confidence Interval for $p_1 - p_2$

Parameters:

- p_1 = In the population of all doctors who received a letter, the proportion who will return the survey
- p_2 = In the population of all doctors who did not receive a letter, the proportion who will return the survey

Contingency Table:

	Return Letter?		Total
	Yes	No	
Receive Letter			
Yes	2570	2448	5018
No	2645	2384	5029

Example - Confidence Interval for $p_1 - p_2$

Sample Statistics:

- $\hat{p}_1 = 2570/5018$
- $\hat{p}_2 = 2645/5029$

$$\begin{aligned} \text{SE}(\hat{p}_1 - \hat{p}_2) &= \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \\ &= \sqrt{\frac{(2570/5018) * (2448/5018)}{5018} + \frac{(2645/5029) * (2384/5029)}{5029}} = 0.009968 \end{aligned}$$

Example - Confidence Interval for $p_1 - p_2$

95% Confidence Interval for $p_1 - p_2$

$$\begin{aligned}\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} * SE(\hat{p}_1 - \hat{p}_2) &= (2570/5018 - 2645/5029) \pm 1.96 * 0.009968 \\ &= (-0.0333, 0.0057)\end{aligned}$$

We are 95% confident in the population of all doctors who received the letter, the proportion who return the survey is between 0.0333 lower to 0.0057 higher than the proportion who return the survey in the population of all doctors who did not receive the letter.